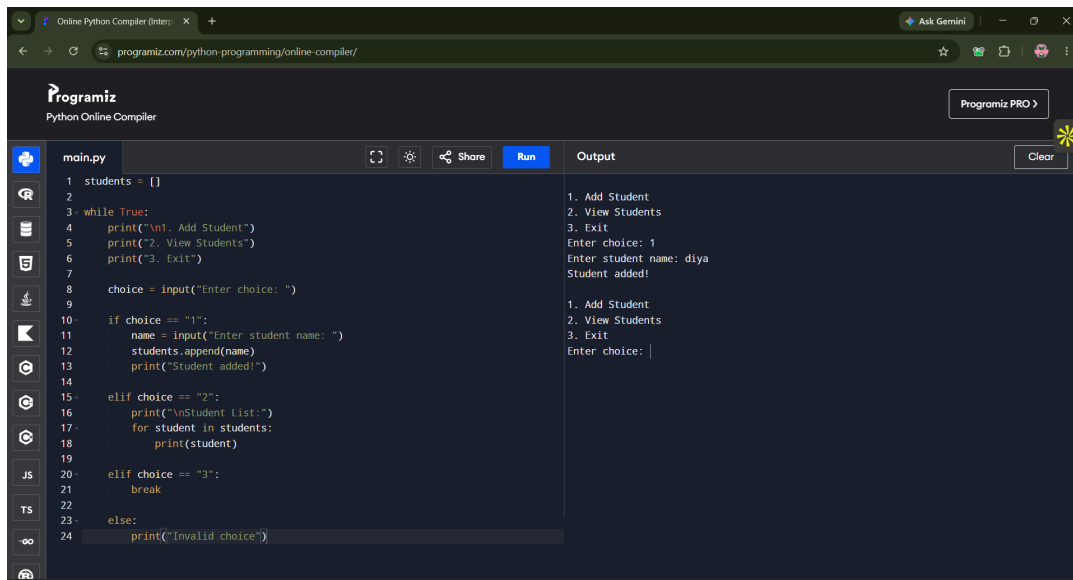


# Lab Assignment 1

## Python Programs

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### 1. Student Management System



The screenshot shows the Programiz Python Online Compiler interface. The code in `main.py` is as follows:

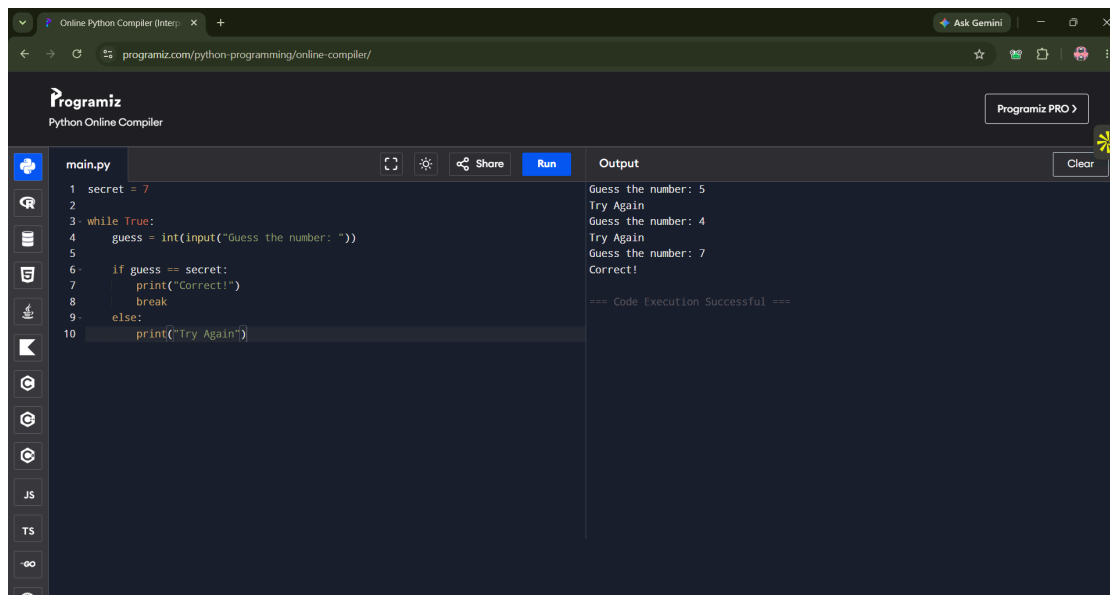
```
1 students = []
2
3 while True:
4     print("\n1. Add Student")
5     print("2. View Students")
6     print("3. Exit")
7
8     choice = input("Enter choice: ")
9
10    if choice == "1":
11        name = input("Enter student name: ")
12        students.append(name)
13        print("Student added!")
14
15    elif choice == "2":
16        print("\nStudent List:")
17        for student in students:
18            print(student)
19
20    elif choice == "3":
21        break
22
23    else:
24        print("Invalid choice")
```

The output shows the program's execution:

```
1. Add Student
2. View Students
3. Exit
Enter choice: 1
Enter student name: diya
Student added!

1. Add Student
2. View Students
3. Exit
Enter choice: |
```

### 2. Guess the number



The screenshot shows the Programiz Python Online Compiler interface. The code in `main.py` is as follows:

```
1 secret = 7
2
3 while True:
4     guess = int(input("Guess the number: "))
5
6     if guess == secret:
7         print("Correct!")
8         break
9     else:
10        print("Try Again")
```

The output shows the program's execution:

```
Guess the number: 5
Try Again
Guess the number: 4
Try Again
Guess the number: 7
Correct!

=== Code Execution Successful ===
```

### 3. Login

The screenshot shows the Programiz Python Online Compiler interface. The code editor on the left contains a Python script for a login system. The output window on the right shows the execution results. The browser address bar shows the URL `programiz.com/python-programming/online-compiler/`.

```
1 username = "admin"
2 password = "1234"
3
4 while True:
5     u = input("Username: ")
6     p = input("Password: ")
7
8     if u == username and p == password:
9         print("Login Successful")
10        break
11    else:
12        print("Wrong Credentials")
```

Output:

```
Username: hello
Password: 789
Wrong Credentials
Username: admin
Password: 1234
Login Successful

=== Code Execution Successful ===
```

## 4. Grocery billing system

The screenshot shows the Programiz Python Online Compiler interface. The code editor on the left contains a Python script for a grocery billing system. The output window on the right shows the execution results. The browser address bar shows the URL `programiz.com/python-programming/online-compiler/`.

```
1 total = 0
2
3 while True:
4     item = input("Enter item (or 'done'): ")
5
6     if item.lower() == "done":
7         break
8
9     price = float(input("Enter price: "))
10    total += price
11
12    print("Total Bill = ", total)
```

Output:

```
Enter item (or 'done'): pen
Enter price: 50
Enter item (or 'done'): pencil
Enter price: 100
Enter item (or 'done'): one
Enter price: 90
Enter item (or 'done'): done
Total Bill = 240.0

=== Code Execution Successful ===
```

## 5. Fibonacci series

The screenshot shows the Programiz Python Online Compiler interface. The code editor contains a Python script named `main.py` that calculates the first `n` terms of the Fibonacci sequence. The code is as follows:

```
1 n = int(input("How many terms? "))
2
3 a, b = 0, 1
4
5 for i in range(n):
6     print(a, end=" ")
7     a, b = b, a + b
```

The output window displays the result of running the code with `n = 5`:

```
How many terms? 5
0 1 1 2 3
=== Code Execution Successful ===
```

## 6. Prime number

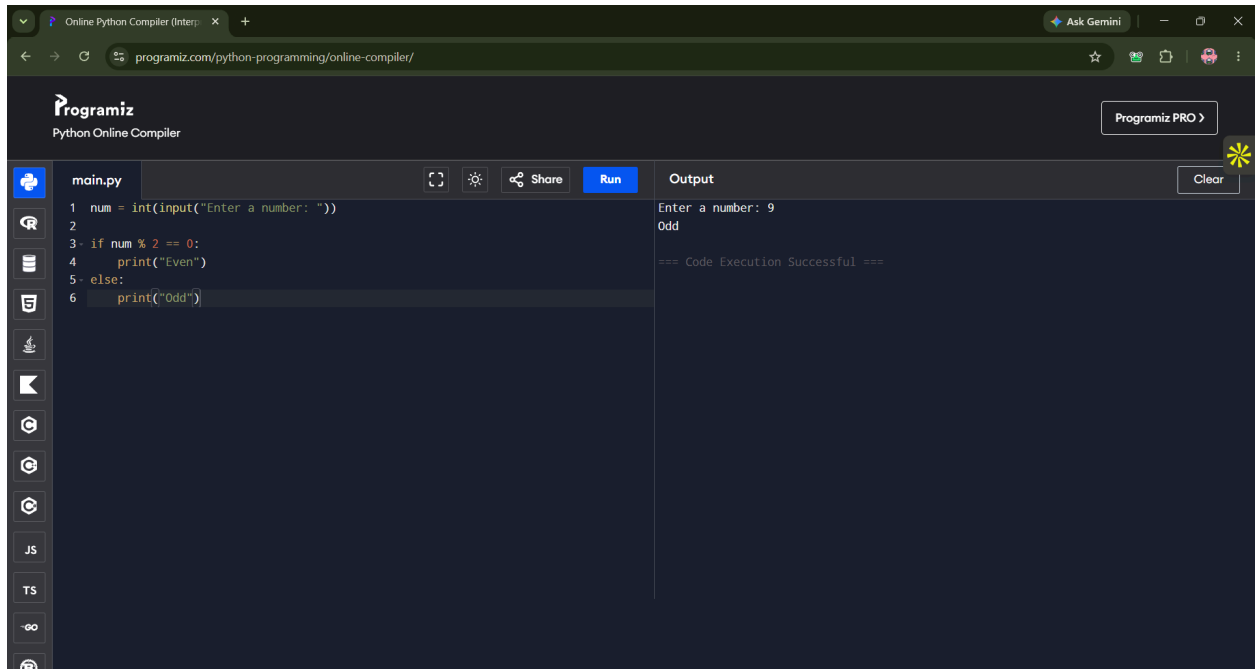
The screenshot shows the Programiz Python Online Compiler interface. The code editor contains a Python script named `main.py` that checks if a given number is prime. The code is as follows:

```
1 num = int(input("Enter a number: "))
2
3 if num <= 1:
4     print("Not Prime")
5 else:
6     for i in range(2, int(num ** 0.5) + 1):
7         if num % i == 0:
8             print("Not Prime")
9             break
10    else:
11        print("Prime")
```

The output window displays the result of running the code with `num = 7`:

```
Enter a number: 7
Prime
=== Code Execution Successful ===
```

## 7. Odd even



The screenshot shows the Programiz Python Online Compiler interface. The code editor on the left contains a Python script named `main.py` that prompts the user to enter a number and checks if it is even or odd. The code is as follows:

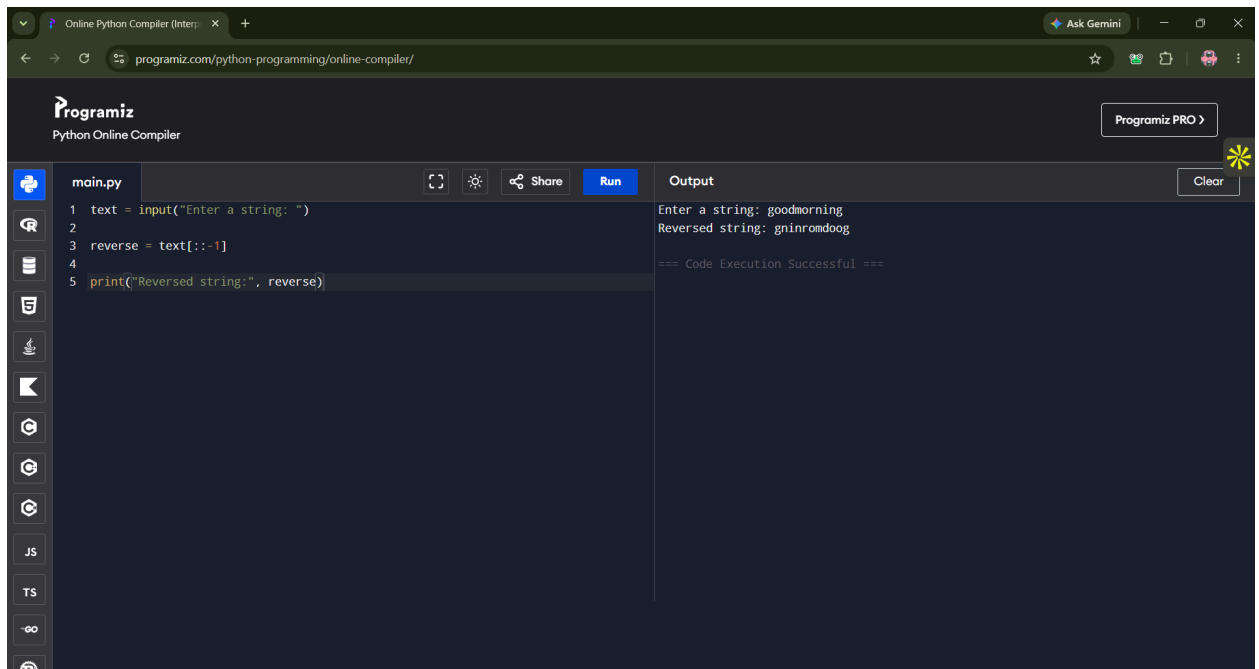
```
1 num = int(input("Enter a number: "))
2
3 if num % 2 == 0:
4     print("Even")
5 else:
6     print("Odd")
```

The `Run` button is highlighted in blue. The output panel on the right shows the result of the execution:

```
Enter a number: 9
Odd
=== Code Execution Successful ===
```

The interface includes a sidebar with icons for file management and a bottom navigation bar with tabs for Python, JS, and TS. A `Programiz PRO` button is located in the top right corner.

## 8. String reversal



The screenshot shows the Programiz Python Online Compiler interface. The code editor on the left contains a Python script named `main.py` that prompts the user to enter a string and reverses it. The code is as follows:

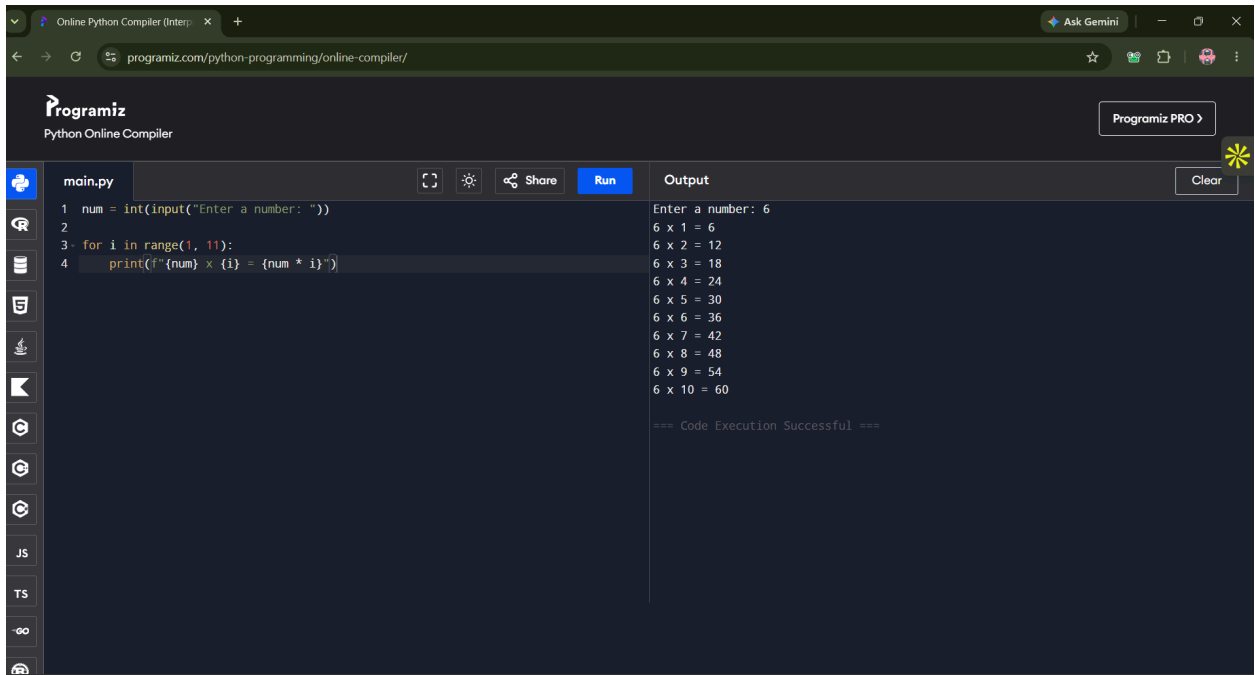
```
1 text = input("Enter a string: ")
2
3 reverse = text[::-1]
4
5 print("Reversed string:", reverse)
```

The `Run` button is highlighted in blue. The output panel on the right shows the result of the execution:

```
Enter a string: goodmorning
Reversed string: gniromdoog
=== Code Execution Successful ===
```

The interface includes a sidebar with icons for file management and a bottom navigation bar with tabs for Python, JS, and TS. A `Programiz PRO` button is located in the top right corner.

## 9. Multiplication table



The screenshot shows the Programiz Python Online Compiler interface. The code editor on the left contains a Python script for generating a multiplication table. The output window on the right displays the result of the program execution, which is a multiplication table for the number 6.

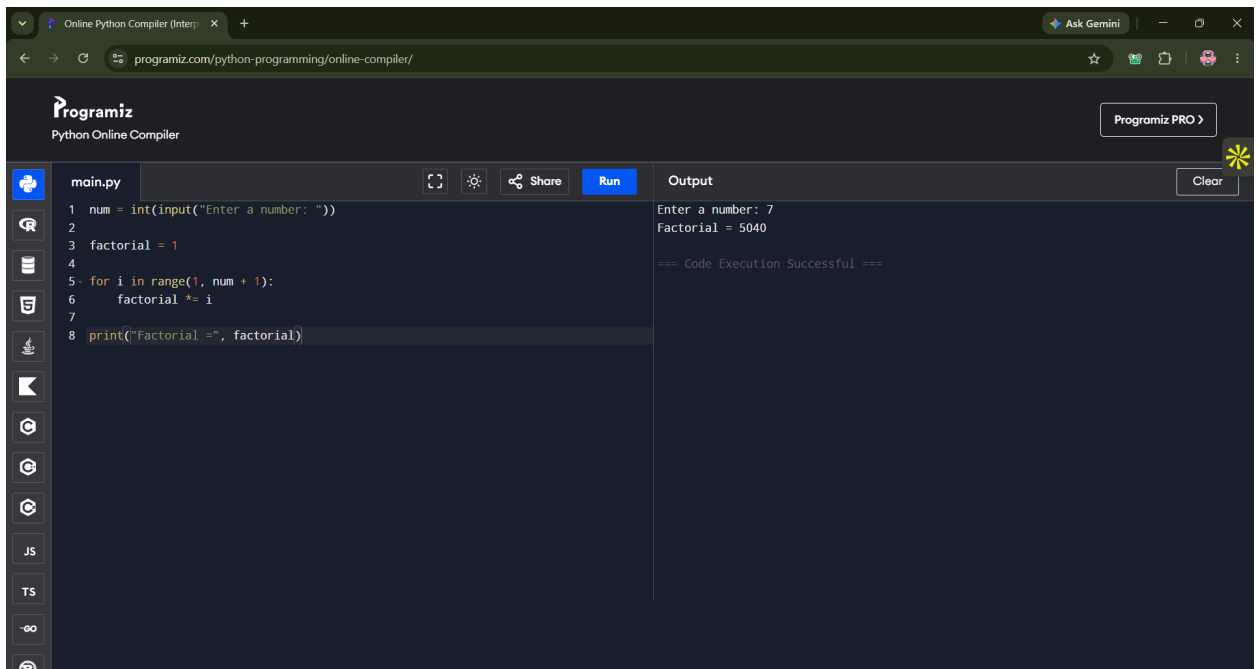
```
main.py
1 num = int(input("Enter a number: "))
2
3 for i in range(1, 11):
4     print(f"{num} x {i} = {num * i}")
```

Output

```
Enter a number: 6
6 x 1 = 6
6 x 2 = 12
6 x 3 = 18
6 x 4 = 24
6 x 5 = 30
6 x 6 = 36
6 x 7 = 42
6 x 8 = 48
6 x 9 = 54
6 x 10 = 60

=== Code Execution Successful ===
```

## 10. Factorial



The screenshot shows the Programiz Python Online Compiler interface. The code editor on the left contains a Python script for calculating the factorial of a number. The output window on the right displays the result of the program execution, which is the factorial of 7.

```
main.py
1 num = int(input("Enter a number: "))
2
3 factorial = 1
4
5 for i in range(1, num + 1):
6     factorial *= i
7
8 print("Factorial =", factorial)
```

Output

```
Enter a number: 7
Factorial = 5040

=== Code Execution Successful ===
```